

Purpose of policy

The aim of this policy is to provide guidance for staff on the indications for isolation precautions and the risk assessment that should be performed prior to decision to isolate.

Isolation precautions should be used for patients who are:

- Either known or suspected to have an infectious disease
- Carrying a Multi-Drug Resistant Organism (MDRO)
- Particularly vulnerable to infection.

The main objectives:

- To ensure minimizing spread of infection to patients and healthcare personnel
- To ensure optimal use of isolation facilities through risk assessment
- To provide the optimal care for infected patients with minimal risk of transmission without compromising their safety

However, standard Infection Prevention control precautions must always be used for all patients regardless of their status (refer to policy on standard precautions). These include the use of gloves, aprons, masks and visors following a risk assessment to identify the risks of exposure to blood, body fluids and microorganisms.

Definition

The term “Isolation” is the use of infection prevention and control precautions aimed at controlling and preventing the spread of infection. There are two types of isolation – Source Isolation (barrier nursing) where the patient is the source of infection and Protective Isolation (reverse barrier nursing) where the patient requires protection i.e. they are immune-compromised.

Scope

This policy includes isolation precautions for both source and protective indications in the Royal hospital at both inpatients, outpatient settings and during transportation of patients. It also includes risk assessment required to assess necessity for isolation. Isolation precautions will be discussed in isolation precaution policy.

Roles & Responsibilities

All staffs working at Royal hospital are responsible for adhering to this policy and for prevention of transmission of infectious diseases by:

1. Treating all bodily fluids as potentially infectious
2. Disposing waste according to hospital policy (refer to policy on management of clinical Waste).
3. Adhering to aseptic technique when required (refer to policy on Aseptic Technique) &Aseptic Technique NUR-CF-SOP-3-Vers.1.7
4. Adhering to hand hygiene practices (refer to policy Hand Hygiene)
5. Maintaining good personal hygiene
6. Observing effective housekeeping practices
7. Adhering to standard precautions (refer to policy on Standard Precautions)

Ward and Department Managers are responsible for ensuring implementation within their area, and that all staffs working within the area adhere to the principles at all times

Bed Managers are responsible for ensuring patients are placed in accordance with this policy, and for escalating any situations where safe placement cannot be achieved. They also should coordinate with infection control practitioners to help with the risk assessment.

Administration is responsible for monitoring implementation of this policy and for ensuring action is taken when staff fails to comply with the policy.

Microbiology laboratory is responsible about detection of infectious agents and reporting results on timely manners to caring team and infection prevention and control.

The Infection Prevention & Control (IP&C)Team is responsible for:

- Providing expert advice in accordance with this policy,
- Supporting staff in its implementation
- Assisting with risk assessment where complex decisions are required.
- Ensuring this policy remains consistent with the evidence-base for safe practice, and for reviewing the policy on a regular basis.

Isolation Facilities

Source isolation is designed to prevent the spread of pathogens from an infected patient to other patients, hospital personnel and visitors.

Source isolation can be achieved by placing patients in:

- 1- **Single rooms** on general wards and multi-specialty ward, with ensuite facilities with or without anteroom.
- 2- **Negative pressure:** These are HEPA- filtered rooms used for high-risk disease/ airborne infection categories, e.g. SARS, Avian influenza, TB, viral hemorrhagic fevers, chickenpox. These rooms are located in the multi-specialty ward and in adult ICU1 and paediatric ICU.
- 3- **Positive pressure rooms** are available in the oncology center that can be used for reverse isolation.
- 4- **Cohort:**
 - When single rooms/isolation rooms are not available and where several patients with the same confirmed organism have been identified, these patients may be placed together in a bay, e.g. Clostridium difficile diarrhea or Methicillin Resistant Staphylococcus Aureus (MRSA). There is a dedicated cohort ward (FM2) to admit all patients colonized or infected with Multi resistant organisms. Please refer to Cohorting Patients for Isolation Purpose Policy (IPC-POL-2-Vers.1.0).
 - Patients tested MDRO positive two or more years ago and not admitted to any hospital after that positive result should not be admitted in cohort ward unless the repeat MDRO screening is still positive.

Criteria for Source Isolation

Patients admitted with the following symptoms must be isolated on admission:

- Known or suspected communicable infection /disease e.g. Pulmonary Tuberculosis, Chicken Pox, viral hemorrhagic fever.
- Unexplained rash if considered to be of an infectious cause
- Multi-Resistant organism's (MDROs) e.g. MRSA, Multi resistant Acinetobacter baumannii (MRAB) and Carbapenem resistant Enterobacteriaceae (CRE).
- ESBL producing coliforms must be isolated in wards that have surgical patients such as surgical wards, maternity wards, gynecology ward, nephrology, urology ward and paediatric surgical ward.
- Patients who are fitting the screening criteria for MDROs should be isolated until the

- screening results are ready (refer to screening policy).
- Diarrhea and/or vomiting until microbiologically proven negative or symptoms subside.
- Suspected viral respiratory illness such as flu, MERS-CoV etc.

If this is not possible because the patient would be at a greater risk by being isolated please contact the Infection Prevention and Control Team for further advice.

Risk Assessment

All patients identified with infectious diseases or alert organisms will be risk assessed for the need for isolation. This will take place between the Infection Prevention and Control Team and the clinical team.

Risk assessment is the assessment of the factors that influence the transmission of a pathogen and its impact. It enables staff to prioritize the use of isolation facilities.

However, the need for isolation of specific infections in side rooms does not take into account the limited resources available, which may lead to inconsistent decisions.

In order to minimize inconsistent decisions, a risk assessment must be undertaken. A discussion **must** take place between the clinical team and Infection Prevention and Control Team.

The following factors will be considered:

- The class of the pathogen and whether there is an effective treatment or vaccine e.g. prioritizing viral hemorrhagic fever over MRSA colonized patient.
- The probable route of transmission and evidence of transmission. Example: prioritize airborne infection such as TB over contact isolation due to Multi resistant organism.
- Susceptibility of the other patients near to the infected patient in the same bay i.e. do the other patients have open wounds or an invasive device. Example isolation of ESBLs in surgical wards and not in medical wards.
- Whether the organism is antibiotic resistant.
- Possible detrimental effects of isolation to the patient i.e. risk of falls, confusion or depression weighed against severity of the risk of transmission to other patients. Isolation should not compromise patient safety and care.

Risk assessment methodology is shown in **Appendix 1**.

Source Isolation Procedure:

- Place an isolation sign outside the door or a reminder sign over the bed space. The isolation sign should be self-explanatory with the type of precautions needed for each case.
- Consider what equipment and supplies are for the area and the patients care.
- Remove all non-essential furniture. The remaining furniture should be easy to clean and should not conceal or retain dirt or moisture either within or around it.
- Ensure that the hand basin has sufficient soap and paper towels for staff use.
- Ensure Alcohol hand gels are full.
- Place yellow clinical waste bag in the room on a foot-operated bin. The bag must be sealed before it is removed from the room. For patients isolated in main bays clinical waste should be placed into a small yellow waste bag and taken to the nearest clinical waste bin
- Keep the patient's personal property to a minimum. All belongings should be washable, cleanable or disposable.

- As far as is reasonably possible provide the patient with his/her own equipment i.e. commodes, sphygmomanometer etc., and all items necessary for attending to personal hygiene. Use disposable items whenever possible. Reusable equipment must be thoroughly decontaminated before being used for another patient.
- Keep dressing solutions, creams and lotions etc., to a minimum and store them within the room. These must be single patient use only
- Personal protective equipment should be kept on a cabinet placed outside the patient room to avoid contamination. Try to keep the minimum required amount. The trolley must be emptied daily and cleaned with Haztab solution.

Entering the room

- Collect all equipment needed.
- Remove any outer clothing and roll up long sleeves to the elbow.
- Put on a disposable plastic yellow apron before entering the room.
- Put on appropriate PPE as per infection control recommendation (refer to Transmission Based Precaution policy)
- Put on disposable gloves only if you are intending to deal with blood, excreta or contaminated material. Gloves do not need to be worn for routine entry into the isolation room.
- Safety glasses, visors or goggles must be available and must be worn when there is a risk of splashing of blood and body fluids.
- Wash hands with soap and water or use Alcohol hand rub before entering the room. N.B. Hand sanitizer Gel must not be used when C.diff or viral diarrhea is known or suspected
- Enter the room, shutting the door behind you.

Attending to the patient

- The patient and relatives must be informed by the care giver about the need for isolation. A full explanation of procedures and precautions must be provided. When needed, the Infection Prevention and Control Team can be contacted to discuss these issues with the patient.
- Ensure the patients are fully aware of the hospital hand hygiene policy and that they feel comfortable to ask staff and visitors to decontaminate their hands if this has not taken place.
- Meals – Whenever possible the patient's meal/water jug should be passed to another member of staff looking after the patient to take into the isolation room. If this is not possible staff serving meals/delivering water jugs should organize the delivery to deliver the meal/jug last. The meal apron can then be disposed of and hands washed on leaving the room.
- The patient must be offered the facilities to decontaminate their hands before eating
- Trays from isolation rooms should be collected last and placed directly onto the trolley. The apron must be removed after the tray is returned and hands MUST be washed. Gloves are NOT required for tray removal.
- Disposable meal trays should be provided to isolated patient.

Excreta.

- Ideally, a toilet should be kept solely for the patient's use. If neither this nor disposable items are available, a separate bedpan or urinal and commode should be left in the patient's room.
- Staff when dealing with excreta must wear gloves and aprons.
- Bedpans and urinals should be covered and taken immediately to the sluice.
- Staff must avoid touching door handles or other surfaces or items. PPE must be removed in the sluice and hands washed before returning to the isolation room where clean PPE should be worn. Commodes must be decontaminated using Haztab solution.

Accidental spills.

Any suspected contaminated fluids must be dealt with immediately according to the hospital policy.

Bathing.

If en-suite facilities are not available an infected patient must be bathed/showered last on the ward. Clean and dry the bath after the previous patient and after the infected patient.

Dressings.

- Aseptic technique must be used for changing all dressings.
- Waste materials and dirty dressings should be discarded in the appropriate yellow clinical waste bag inside the room.
- Used lotions, creams, etc., must be kept in the room and not used for other patients).
- Sterile packs must be stored safely to protect them from contamination and damage. Please refer to hospital aseptic technique policy.

Linen.

- Place infected linen in a red alginate polythene bag, which must be secured tightly before it leaves the room.
- Just outside the room, place this bag into a red linen bag, which must be secured tightly and not used for other patients. These bags should await the laundry collection in a safe area.

Waste:

Yellow clinical waste bags should be kept in the isolation room for disposal of all waste generated in the isolation room. The bag should be sealed well before leaving the room.

Leaving the room

- If wearing gloves, remove and discard them in the yellow clinical waste bag.
- Remove apron and discard it in the appropriate bag. Wash hands or use Alcohol hand rub
- Leave the room, shutting the door behind you.
- Wash hands with Soap and water or rub hands with alcohol hand rub.

Daily Cleaning of the Room

- NB. It is of paramount importance that the room or bay or bed space is cleaned daily. Housekeeping staff must be given access to the room or bay and ward staff must facilitate this process by working with housekeeping services to achieve high standards of cleanliness.
- Housekeeping staff must understand why isolation is required and should be

instructed on the correct procedure. Nursing staff must work in close collaboration with the cleaners to ensure correct procedures are followed.

- The area where isolation is being carried out must be cleaned last.
- Separate cleaning equipment must be kept for this area. The yellow color-coding system must be used when cleaning isolation rooms, bay or bed spaces. This will include mops and buckets. Refer to cleaning/terminal cleaning policies.
- Cleaners must wear PPE while cleaning and handling waste within the isolation room/bay or bed space.
- Disinfectant for daily cleaning: (disinfectant detergent) to be utilized for all isolation rooms except rooms occupied by C.difficile positive and VHF cases.
- Floor must be disinfected as above.
- Buckets should be filled in the domestic room or sluice they must not be filled from the clinical hand basin. Cleaning solutions must be changed after each room or bed space as a minimum.
- Cleaning solutions/disinfectant must be freshly diluted to the correct dilution following the manufacturer's instructions. Under no circumstances should products other than those agreed by Infection Prevention and Control and the housekeeping services be used.
- After use, the bucket must be washed and dried and returned to the cleaner room.
- Mop heads if not disposable should be removed after use with isolation patients and laundered in a hot wash after cleaning.
- Furniture and fittings should be damp dusted using disposable J cloths with the appropriate disinfectant solution.
- The toilet, shower and bathroom area must be cleaned at least once a day. Cleaning items must have red color-coding.
- Cloths must be disposed of in the clinical waste bin within the room

Transporting Infected Patients outside the Isolation Area to Other Departments:

- Patient with infectious conditions should only leave the ward for essential reasons.
- Inform the receiving department concerned about the diagnosis. Good communication is vital to prevent risks to others.
- Arrange for the patient to have the last appointment of the day whenever possible and ensure the receiving department are aware of the patient's impending arrival
- Any porters involved must be instructed carefully on any precautions required ensuring that the patient confidentiality is maintained.
- Porter staff entering the isolation room should put on appropriate PPE. The PPE should be removed once the patient is settled in the chair/bed or trolley and hands must be washed or decontaminated using alcohol gel on leaving the room.
- Gowns and gloves must NOT be worn to transfer patients through the hospital except in patients with viral hemorrhagic fever (VHF) and MERS Corona virus(MERS CoV), COVID 19 and for ICU patients if direct care needed to be delivered during transport.
- In some circumstances, for example tuberculosis, the patient should wear a surgical mask when leaving the room.
- On entering the department the patient must be taken straight to the procedures room and must not wait in the general waiting area.
- All staff attending the patient must wear appropriate PPE to move or handle the isolation patient while they are in the department.
- 5 moments of Hand hygiene should be always maintained.

- All equipment used in the room must be cleaned thoroughly after the patient has left the department. The trolley or chair should be cleaned after use. Disinfection wipes will be sufficient for this process.

Transportation of specimens of VHF cases:

- All specimens must be treated as biohazard
- Label with biohazard label
- Mark request form accordingly
- Double bag
- For confirmed cases all specimens will be handled at Containment Level 3.
- Transportation of samples between laboratories should be by enhanced category B (category 'A' packaging and category 'B' courier).

Transfer of isolated patients to other institutions:

- Inform the receiving hospital (infection control or bed manager) about the case and the isolation precautions necessary.
- Transfer of highly infectious cases such as MERS CoV and VHF cases to another hospital should be avoided unless it is necessary for medical care.
- If transfer is essential, the Infection Control Team at the receiving hospital and the ambulance staff must be advised in advance of the special circumstances of the transfer.
- Appropriate Personal Protective Equipment must be worn during transferring the patient in the ambulance

Discharging the Patient – Terminal Cleaning of the Room

- Nursing staff should inform the bed manager when the patient is due for transfer to another hospital to inform the hospital to provide the appropriate bed.
- Nursing staff must inform the disinfection team that the room will need to be terminally cleaned. Hydrogen peroxide vapor machine should be used as much as possible.
- Curtains should be changed before cleaning
- Nursing staff should strip the bed and clean all medical and nursing equipment with the appropriate disinfectant solution. This should then be removed from the room by nursing staff to allow the disinfection team staff to clean.
- Impervious surfaces, e.g. Tables and chairs, window sills etc., should be cleaned and disinfected thoroughly.
- Patients' lockers must be cleaned thoroughly including the inside.
- The floor must be disinfected and dried thoroughly. The bed should be pulled out to ensure the space behind the bed is cleaned thoroughly.
- The room can be reused as soon as it has been correctly and thoroughly cleaned. Allow room to dry thoroughly before use.
- For more details please refer to Terminal cleaning procedure (IPC-SOP-1-Vers.1.0)

Curtains

Curtains must be changed on termination of isolation and every two weeks if isolation continues longer than 2 weeks.

The contaminated curtains should be removed before the room is cleaned. The clean curtains

should be hung after the room has been cleaned.

Protective Isolation

Protective isolation is intended to prevent a more susceptible patient acquiring infection e.g. patients with lowered immune systems.

- This is best achieved in a positive pressure side room in oncology ward, although a general side room may be used with the door remaining closed.
- Staff/ visitors with infections including colds, flu like symptoms and active cold sores should not care for or visit the patient.
- The room/ furniture should be cleaned with detergent and water prior to admitting the patient.
- Staff must ensure that they decontaminate their hands and put on gown, gloves and mask before entering the room.
- Dustbins should be lined with black bag if the case is purely for protective isolation.

Monitoring Arrangements

Regular Monitoring

The policy will be monitored using isolation audit tool. Results will be reported to the Infection Control Committee who will review compliance and if necessary nominate a manager to develop an action plan to achieve the standards / process set out in the document. This action plan will also be monitored via IP&C Committee.

Appendix 1

Alert organism risk assessment for the use of isolation rooms

All patients suspected or known to be colonized or infected with an infectious disease or condition must be isolated, in line with Isolation Policy. However, due to limited isolation facilities it is recognized that, at times, single rooms will need to be prioritized. The following information is intended to assist Senior Nursing staff, Bed Management teams and Ward Managers in this process. For the further list of infectious diseases and the isolation precaution needed please refer to GCC infection control manual in the Royal portal pages 45-58.

Code	Action
3	High priority for a single room or isolation room with negative pressure. Inform Infection Prevention and Control Team(ICT) if single room or isolation room not available.
2	Single room required. Assess patients currently in side rooms. If single room not available, contact Infection Prevention and Control to discuss further. Where side room n/a, nurse in main bay, providing other patients in the bay are not deemed vulnerable i.e. with open wounds or invasive devices, immune-compromised etc. Move to single room ASAP.
1	Low risk. Single room or cohort bay required, but may be nursed in bay providing other patients in the bay are not deemed vulnerable i.e. with open wounds or invasive devices, immune-compromised etc
0	Does not require Isolation
*	May require contact tracing

Risk assessment for isolation of Multi-drug resistant organisms and some selected common infections.

Alert organism/assessment	Precautions	Level	Comments
Multi Drug Resistant Gram negative MDRAB, CRE (MDRO) Colonized in screening body sites i.e. nose, invasive devices, i.e. PVC Positive Sputum Oozing wound Tracheostomy	Contact	2*	Priority to go to isolation ward All Gram negative MDROs must be cohorted in the isolation ward if fitting the criteria(refer to cohort policy)
Other Multi resistant organisms i.e. ESBL Catheterized Oozing wound	Contact	2*	Isolate only those who are admitted in surgical wards or admitted near surgical patients Don't isolate cases in medical wards
MRSA Skin colonization	Contact	1*	
MRSA Positive Wound pouring pus Sputum with cough or tracheostomy	Contact	2*	Prioritize MRSA to go to single rooms or if MRSA cohort is available can be cohorted
High risk patients awaiting body screening for MDROs As per screening policy	Contact	2*	Apply precautions until all results are available
Diarrheal/ vomiting illness suspected to be infectious C.difficile, Campylobacter Norovirus etc.	Contact enteric precautions	2	Vomiting should precedes diarrhea for isolation if single rooms are limited
Chickenpox Rash developed within previous 10 days	Airborne and contact	3*	Only staff with a history of Chicken Pox should have contact with patient

			Isolate until all lesions are crusted
Chickenpox Rash dry, longer than 10 days old	Standard	0	
Shingles Rash in an exposed area Such as face	Airborne and contact	3*	Isolate until lesions are crusted
Shingles Rash is disseminated to 3 more dermatomes	Airborne and contact	3*	Isolate until lesions are crusted
Shingles Localized in immune-compromised	Airborne and contact	3*	Until disseminated zoster is ruled out, then can revert to standard precautions
Shingles Localized and covered in immune-competent	Standard	0	Cover lesions, wear gloves if contact with lesions
Confirmed pulmonary TB	Airborne	3*	Until 14 days of treatment With 3 negative AFB smears and clinical /radiological improvement
Suspected Pulmonary TB	Airborne	3*	Isolate until three AFB sputum samples are negative on microscopy If low risk 1 negative TB PCR(ultrasensitive) If high risk 2 negative PCR or send more samples for microscopy.
Suspected or confirmed MDR Pulmonary TB	Airborne	3*	Requires negative pressure ventilation in multi-specialty ward (priority), consult infection control for de-isolation
Extra-pulmonary TB Aerosolizing procedures (surgery) or draining wounds Inter-costal drain	Airborne	3*	
Extra-pulmonary TB Without risk of aerosol	Standard	0	
Scabies	Contact precautions	2*	
Group A Strep	Droplet/contact	2	Until 48 hours of appropriate antibiotics

Head Lice	Contact precautions	2	Clothing and linen used by the infected patient from 3 days prior and 24 hours after treatment must be placed in a hot water soluble bag labeled and sent to laundry.
Suspected meningococcal Meningitis	Droplet	3*	Until 24 hours of antibiotics. Inform Infection Prevention and Control if indirect contact with respiratory secretions during resuscitation
Influenza / other viral respiratory infections	Droplet	2*	Until 4 days after start of Flu treatment or until clear of symptoms (for critical care or immune-compromised patient discuss with infection control)
Measles	Airborne	3*	Isolate until 4 days after appearance of the rash For duration of illness in immune-compromised Assign immune HCWs to take care of measles cases
Mumps	Droplet	2*	Until 9 days from appearance of parotitis
Rubella	Droplet	2*	Until 7 days after onset of rash
MERS corona	Airborne	3*	Isolate for the duration of illness
Viral hemorrhagic fever	Airborne/contact precautions	3*	Inform infection Prevention and Control or Microbiologist on call immediately if diagnosis is suspected

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
GCC Infection Prevention & Control Manual	GCC center for infection control	2013	58-68
Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Healthcare Settings	CDC	2007	18, 71-72
Heart of England NHS Foundation Trust Isolation Policy	Trust's Infection Prevention and Control Team	2010	1-39

Review / Revision History

Revision History			
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